40. Find $E(3X)$ and $E(4Y)$ for the following joint probability distribution function of X and Y

X \ Y	1	2	3
−5	0	0.1	0.1
0	0.1	0.2	0.2
5	0.2	0.1	0

Answers:

I. 1. *c*. 2. *c* 3. *b* 4. *a* 5. *a* 6. *b* 7. *c* 8. *a*

II. 9. random variable 10. 0 11. $F(b) - F(a)$ 12. 0 13. $f(x)$

14. $E(X)$ 15. arithmetic mean

III. 21. $\frac{1}{28}$ 22. 1 23. $\frac{1}{2}$

IV. 25.

X	0	1	2	3
$P(x)$	$\frac{125}{216}$	$\frac{75}{216}$	$\frac{15}{216}$	$\frac{1}{216}$

26.

X	0	1	2	3
$P(x)$	$\frac{5}{30}$	$\frac{15}{30}$	$\frac{9}{30}$	$\frac{1}{30}$

27. $\frac{3}{4}, \frac{1}{4}$ 28. 5

V. 29. $1, \frac{1}{5}$ 30. $6, \frac{3}{8}$ 31. 7 32. 7

33. $\frac{3}{2}, \frac{3}{4}$ 34. $\frac{1}{81}, \frac{1}{9}, \frac{56}{81}, \frac{11}{27}$

35. $\frac{2}{13}, \frac{24}{169}$ 36. −15 37. $\frac{3}{4}$

38. (i)

X	0	1	2	3
$P(x)$	$\frac{64}{343}$	$\frac{144}{343}$	$\frac{108}{343}$	$\frac{27}{343}$

(ii)

X	0	1	2	3
$P(x)$	$\frac{5}{30}$	$\frac{15}{30}$	$\frac{9}{30}$	$\frac{1}{30}$

39. 0.2, 8.7 40. −1.5, 2.0